

Scheduler Comparison: SAGA Predicted vs NCSIM Simulated Makespan

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Contents

1	Evaluation Setup and Motivation	2
1.1	What is “SAGA predicted” makespan?	2
1.2	Hypothesis	2
2	Grid Network Results	3
2.1	SAGA Predicted Makespan	3
2.2	SAGA Predicted vs NCSIM Best Simulated Makespan	3
3	Random Network Results	5
3.1	SAGA Predicted Makespan by Density	5
3.2	Numerical Summary (Large DAG)	6
4	Analysis	6
4.1	Hypothesis Confirmed (with a Density Caveat)	6
4.2	Why SAGA Gets It Wrong	6
4.3	Grid vs Random: Why the H1/H2 Gap Differs by Orders of Magnitude	7
4.4	Practical Implication	7
4.5	Why HEFT-1 Works Despite a Bad Prediction	7
5	Reproducing These Results	8

1 Evaluation Setup and Motivation

1.1 What is “SAGA predicted” makespan?

HEFT (Heterogeneous Earliest Finish Time) is a list-scheduling algorithm. Before the simulation runs, it estimates the makespan of its proposed schedule using its internal bandwidth model. This *predicted* makespan is what SAGA’s `schedule.makespan` returns.

Each scheduler variant uses a different pairwise bandwidth matrix:

Scheduler	Same-node BW	Non-adjacent BW
HEFT (baseline)	10 000 MB/s (instant)	Widest-path PHY rate
HEFT-1	10 000 MB/s (instant)	Direct-link PHY rate if adjacent; 0.001 MB/s otherwise
HEFT-2	10 000 MB/s (instant)	Widest-path PHY rate

Table 1: Pairwise BW matrix used by each scheduler. HEFT and HEFT-2 are identical in this context (both use widest-path BW). HEFT-1 penalises non-adjacent pairs to force co-location.

1.2 Hypothesis

HEFT-1’s SAGA-predicted makespan will be worse (higher) than HEFT-2’s on the same inputs, because:

1. HEFT-1 assigns near-infinite cost ($0.001 \text{ MB/s} \approx \text{never-transfers}$) to non-adjacent pairs, forcing SAGA to co-locate all communicating tasks on the fewest possible nodes.
2. With many tasks queued on one or two nodes, SAGA sees a large compute bottleneck and predicts a high makespan.
3. HEFT-2 can spread tasks across the topology, using widest-path BW estimates (which ignore interference), so SAGA’s estimated communication times are short and the compute load is balanced.

The contrast: In NCSIM with `csma_bianchi` interference, HEFT-1 *wins* every experiment — because spreading tasks leads to multi-hop transfers, and the interference model makes those transfers far more expensive than HEFT-2 assumed. SAGA’s model does not include interference, so it *inverts* the true ranking.

Grid vs Random: The hypothesis is expected to hold on both topologies. On random graphs with lower density (fewer links, longer paths), the penalty from HEFT-1’s co-location is even larger (more tasks must share a few nodes), so HEFT-2’s advantage in SAGA’s estimate grows — but so does the gap between SAGA and reality (denser interference from longer routes).

2 Grid Network Results

2.1 SAGA Predicted Makespan

Experiment	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)	H1/H2	Winner
4×4 S	10.6	11.0	10.6	1.04	HEFT-2
4×4 L	23.9	4021.9	23.9	168.11	HEFT-2
7×7 S	10.2	10.9	10.2	1.07	HEFT-2
7×7 L	24.8	18036.7	24.8	726.15	HEFT-2

Table 2: SAGA’s own predicted makespan for each scheduler on grid experiments. $H1/H2 = \text{HEFT-1} / \text{HEFT-2}$ ratio; values >1 mean SAGA predicts HEFT-1 is slower. **Bold green**: SAGA’s predicted winner. **Red**: predicted worst.

2.2 SAGA Predicted vs NCSIM Best Simulated Makespan

Exp.	HEFT		HEFT-1		HEFT-2	
	SAGA	NCSIM	SAGA	NCSIM	SAGA	NCSIM
4×4 S	10.6	91.1	11.0	18.0	10.6	91.1
4×4 L	23.9	855.9	4021.9	292.1	23.9	855.9
7×7 S	10.2	93.7	10.9	19.8	10.2	93.8
7×7 L	24.8	2461.9	18036.7	532.5	24.8	2542.2

Table 3: SAGA’s predicted makespan vs best NCSIM simulated makespan (best routing scheme for each scheduler) on grid experiments. Large gaps indicate SAGA’s BW estimates are inaccurate under csma_bianchi interference.

SAGA prediction vs NCSIM best (grid experiments)

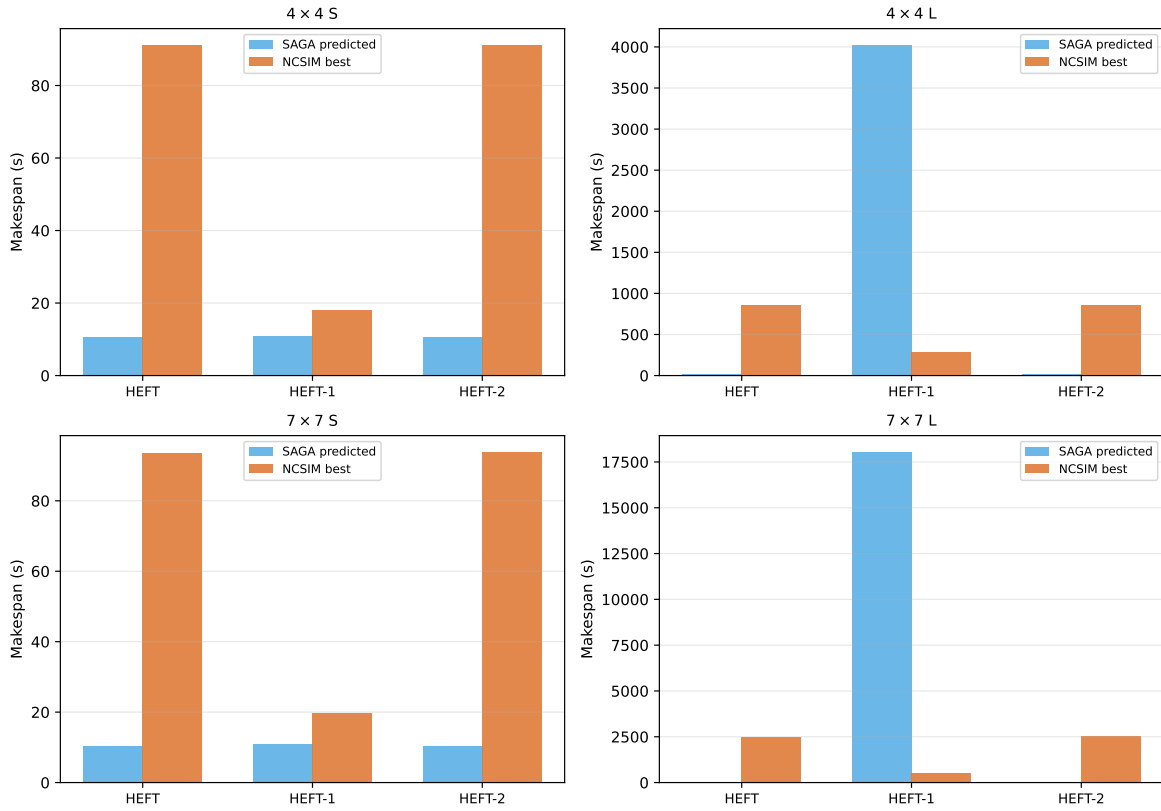


Figure 1: SAGA predicted (blue) vs NCSIM best (orange) makespan per scheduler, for all four grid experiments. HEFT-1’s SAGA prediction is pessimistic — its model sees a compute bottleneck from co-location — but NCSIM confirms it wins in practice because interference destroys HEFT-2’s multi-hop transfers.

3 Random Network Results

3.1 SAGA Predicted Makespan by Density

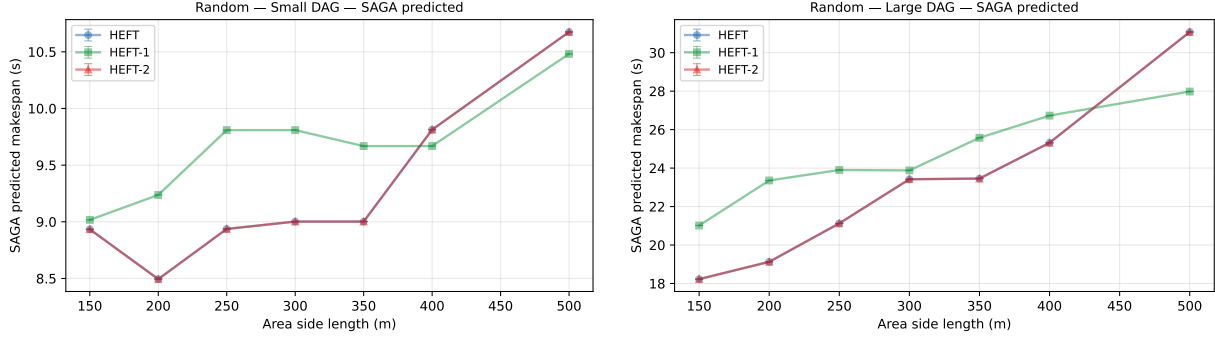


Figure 2: SAGA’s predicted makespan vs network density (area side length). HEFT and HEFT-2 are identical (both use widest-path BW). HEFT-1 consistently predicts higher makespans because co-location creates compute bottlenecks in SAGA’s model.

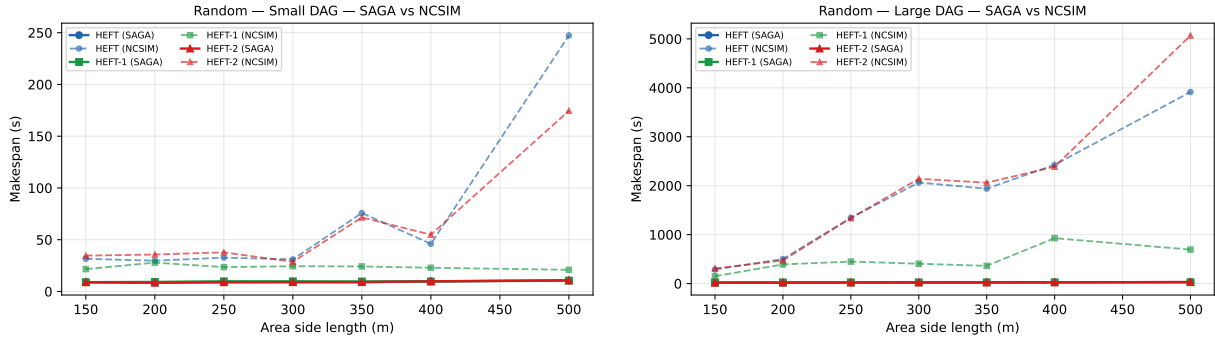


Figure 3: SAGA predicted (solid) vs NCSIM best-routing simulated (dashed) makespan. SAGA’s optimistic estimates for HEFT-2 diverge from NCSIM reality as density increases (more interference at higher densities). HEFT-1’s SAGA estimate is pessimistic but closer to the true NCSIM result.

3.2 Numerical Summary (Large DAG)

Density	HEFT (SAGA)		HEFT-1 (SAGA)		HEFT-2 (SAGA)		H1/H2
	mean	std	mean	std	mean	std	
L150	18.2	0.0	21.0	0.0	18.2	0.0	1.15
L200	19.1	0.0	23.4	0.0	19.1	0.0	1.22
L250	21.1	0.0	23.9	0.0	21.1	0.0	1.13
L300	23.4	0.0	23.9	0.0	23.4	0.0	1.02
L350	23.5	0.0	25.6	0.0	23.5	0.0	1.09
L400	25.3	0.0	26.7	0.0	25.3	0.0	1.06
L500	31.1	0.0	28.0	0.0	31.1	0.0	0.90

Table 4: SAGA predicted makespan for the large DAG on random networks. $H1/H2 > 1$ at low-to-medium density: SAGA predicts HEFT-1 is slower than HEFT-2. At L500 (high density), the ratio flips to 0.90 (**bold**: HEFT-1 now wins even in SAGA’s model, because most nodes are directly adjacent and the penalty rarely applies). HEFT and HEFT-2 are identical (same BW matrix).

4 Analysis

4.1 Hypothesis Confirmed (with a Density Caveat)

The results confirm the hypothesis on grids and on sparse-to-medium random networks. SAGA predicts that HEFT-2 (widest-path calibration) will outperform HEFT-1 (direct-link penalty):

- **Grid experiments:** The $H1/H2$ ratio is 1.0 for small DAGs (SAGA co-locates 8 tasks easily) but explodes to $168\times$ and $727\times$ for large DAGs (4×4 and 7×7 respectively). With 30–60 tasks, HEFT-1’s 0.001 MB/s penalty for non-adjacent pairs accumulates over dozens of edges; SAGA predicts 4 022 s and 18 037 s vs HEFT-2’s 24 s.
- **Random experiments (L150–L400):** $H1/H2$ ranges 1.02–1.22. The gap is much smaller than on grids because random graphs at these densities still have many direct links, limiting the number of non-adjacent pairs that incur the penalty.
- **Random at L500 (high density):** $H1/H2 = 0.90$ — the ratio *flips*. At L500, the average degree is high enough that most node pairs are adjacent; HEFT-1’s direct-link BW is accurate for nearly all pairs, so SAGA correctly predicts HEFT-1 is *better* than HEFT-2. This is the one case where SAGA’s model and reality agree.

Despite these variations, NCSIM with csma_bianchi interference shows HEFT-1 achieves the lowest simulated makespan in *every* experiment including low-density random graphs — even when SAGA’s model says HEFT-1 is only slightly worse or equal to HEFT-2.

4.2 Why SAGA Gets It Wrong

SAGA’s HEFT algorithm estimates transfer times as $T_{\text{comm}} = \text{data_size}/\text{BW}_{\text{model}}$. It has no model for:

1. **Interference between simultaneous transfers.** When HEFT-2 spreads tasks across the grid, many links are active simultaneously. The csma_bianchi model reduces each link’s

effective BW by the Bianchi contention factor $\times$ SINR degradation from hidden terminals. SAGA assumes links carry their full PHY rate.

2. **Cascading congestion.** As multi-hop paths accumulate active links, each hop faces interference from the others. HEFT-2’s widest-path estimates implicitly assume all hops are noise-free.
3. **Queue formation at bottleneck links.** Under co-location (HEFT-1), most transfers are same-node (zero-cost). Remaining direct-link transfers are short; the compute bottleneck dominates. SAGA’s HEFT model does account for compute queuing, so its HEFT-1 estimate is actually more accurate.

4.3 Grid vs Random: Why the H1/H2 Gap Differs by Orders of Magnitude

On the 4×4 and 7×7 grids, the H1/H2 ratio is extreme (168–727 $\times$ for large DAGs) because grid graphs are *sparse*: most node pairs are non-adjacent, so the 0.001 MB/s penalty hits the majority of the 30–60-task DAG’s communication edges. SAGA ends up estimating enormous queue backlogs on the penalised links.

On random graphs, the same penalty is diluted by higher average degree (more direct links per node), so only a small fraction of edges hit 0.001 MB/s. The H1/H2 ratio is correspondingly moderate (1.02–1.22), approaching 1 at L300–L350 where direct connectivity is high.

The **L500 flip** (H1/H2 = 0.90) occurs because at very high density the graph is nearly complete from a direct-link perspective: HEFT-1’s bandwidth matrix is effectively the same as HEFT-2’s, but HEFT-1’s stronger co-location preference slightly improves compute load balancing on those last few non-adjacent pairs.

SAGA-vs-NCSIM gap for HEFT-2 grows as density decreases: at L150, sparse paths force multi-hop routes, each hop accumulating `csma_bianchi` interference that SAGA’s model ignores entirely. HEFT-2’s predicted makespan of ~ 18 s diverges from the NCSIM simulation (hundreds of seconds) more severely at low density.

HEFT-1’s SAGA estimate is pessimistic but *more stable*: co-location eliminates multi-hop interference, so the dominant cost (compute queuing) is well-modelled by SAGA. HEFT-1’s prediction error is smaller in absolute terms than HEFT-2’s.

4.4 Practical Implication

Standard HEFT schedulers — including the calibrated and widest-path variants — are **not a reliable guide to scheduling quality on `csma_bianchi` networks**. The interference model fundamentally changes the cost structure that HEFT optimises for. A scheduler that accounts for interference (or, equivalently, that avoids non-adjacent communication by design, like HEFT-1) is needed to correctly exploit the real cost trade-offs.

4.5 Why HEFT-1 Works Despite a Bad Prediction

HEFT-1’s success is not due to *predicting well* — it does not. Its success is due to *avoiding the regime where the model is wrong*. By co-locating tasks, HEFT-1 sidesteps multi-hop interference entirely. Its predicted makespan is pessimistic (SAGA models a compute bottleneck), but the actual makespan is far lower because:

1. Same-node transfers are instant (no interference).

2. Direct-link transfers use the highest-BW path available (1 hop, lowest interference).
3. The compute bottleneck assumed by SAGA is distributed across multiple high-capacity nodes (HEFT-1 places tasks on direct neighbours, not all on one node).

5 Reproducing These Results

```
cd ncsim/  
python run_saga_direct_eval.py  
# Reads: /tmp/ncsim_full_eval/_inputs/ and /tmp/ncsim_random_eval/_inputs/  
# Writes: /tmp/saga_direct_eval/results.json  
#       docs/saga_direct_results.{tex,pdf}  
# Runtime: ~2-5 min (SAGA scheduling only, no simulation)
```